
R PAPER .docx

Thesis citation audit

GENERATED

3 September 2026

JUDGED BY

not recorded

PROMPT VERSION

citation_support-v1.4

SENTENCES READ

84

SENTENCES CHECKED

79

Every verdict in this report was produced on this machine by not recorded (citation_support-v1.4). No manuscript text was sent anywhere.

Summary

84 sentences were read and 79 were checked against a source or judged for whether they need one.

- 65 citation need
- 19 unverifiable

What the model concluded:

- 41 needs citation
- 19 no citation needed

Claims checked against their source

Each sentence below cites a source Gaply could read. The quoted passage is what the verdict rests on — read it before the verdict. None found.

Sentences that may need a citation

These sentences carry no citation. Whether they need one is a judgement, not a fact, and it is the model's.

Sentence 0 · page unknown

“2Department of Computer Science DPG Degree College, Gurugram, India”

Verdict: no citation needed

Reason: This is a general statement about the department and college, not requiring citation.

Sentence 1 · page unknown

“This paper presents a bidirectional long-term short-term memory (BiLSTM) network-augmented attention pooling with a novel end-to-end framework called HEFCSO-BiLSTM in which an attention-based optimization hybrid model, HEFCSO, consisting of the global FA and the local CSA exploration-exploitation framework, adapts the BiLSTM weights, biases, dropout rates, and the learning rate to minimize categorical cross-entropy loss.”

Verdict: needs citation

Reason: This is an empirical claim about a novel end-to-end framework (HEFCSO-BiLSTM) and a new model (BiLSTM network-augmented attention pooling) which are not common knowledge and require citation.

Sentence 2 · page unknown

“Combining domain-specific preprocessing, 200-dimensional GloVe tweet embeddings, and SMOTE-Text class balancing for end-to-end text processing, the model is extensively analyzed for this study in which three curated datasets crescent up to old benchmark corpora of SemEval-2018, ISEAR, and GoEmotions.”

Verdict: needs citation

Reason: The sentence contains empirical claims about the model's performance and the datasets used, which require citation.

Sentence 3 · page unknown

“In the combined data set, the results achieved 96.42% accuracy, 95.00% F1-score, and an MCC of 0.945 for SemEval 2018, with 10 percentage points better than the best baseline in accuracy.”

Verdict: needs citation

Reason: The sentence contains specific statistical results (accuracy, F1-score, MCC) from a particular study (SemEval 2018). These results are considered empirical and require citation.

Sentence 4 · page unknown

“We conducted ablation studies, statistical significance ($p < 0.05$), and an analysis of the interpretability of the weights of the attention.”

Verdict: needs citation

Reason: The sentence contains empirical claims (ablation studies, statistical significance) and methodological details (analysis of interpretability of attention weights) which require citation.

Sentence 5 · page unknown

“The meteoric rise of user-generated content on platforms like Twitter, Reddit, or Instagram has made social media an instantaneous socioemotional gauge.”

Verdict: no citation needed

Reason: This is a general observation about social media trends and does not require citation.

Sentence 7 · page unknown

“Despite the good progress made so far, revolutionizing the words in social media text is still non-trivial, interrupted by the informal spellings, sarcasm, emoji-containing expressions, huge class imbalances, and small posts.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce new information that requires citation. It continues the idea of progress made so far.

Sentence 9 · page unknown

“Deep learning definitely comes in handy in learning Columbus's 1492.”

Verdict: needs citation

Reason: The sentence makes an empirical claim about deep learning being useful, which is not common knowledge and requires citation.

Sentence 11 · page unknown

“However, BiLSTM-like architectures, as high-performing as they are, are very sensitive to hyper parameterization; even a slightly poorer configuration may lead to a drop in performance by more than 10 percentage points on our ablation experiments.”

Verdict: needs citation

Reason: The sentence makes an empirical claim about the performance sensitivity of BiLSTM-like architectures, which requires citation as it is an observation about the model's behavior.

Sentence 12 · page unknown

“Automatically searching through various hyperparameter spaces is a rather complex and resource-intensive process.”

Verdict: no citation needed

Reason: This is a general statement about the nature of hyperparameter search, which is considered common knowledge in the field of machine learning.

Sentence 13 · page unknown

“The Firefly Algorithm (FA) is especially good at accelerating worldwide exploration, whereas the Crow Search Algorithm (CSA) works to break up local areas effectively.”

Verdict: no citation needed

Reason: The sentence is a general statement about the capabilities of FA and CSA, which is common knowledge.

Sentence 14 · page unknown

“However, neither strategy of global nor local path exploration is ever enough to satisfy complex optimization.”

Verdict: needs citation

Reason: This is an empirical claim about optimization strategies which is not the author's own finding and is not common knowledge.

Sentence 15 · page unknown

“Hence, this paper introduces the Hybrid Emotional Firefly–Crow Search Optimization (HEFCSO) algorithm, which executively switches from global to neighborhood searches under dynamic stimulus action.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce a new claim or result.

Sentence 16 · page unknown

“The complete-flow architecture is depicted in Fig. 1.”

Verdict: no citation needed

Reason: This is a general statement about the architecture and does not require citation.

Sentence 17 · page unknown

“The research contributions of the study are outlined as follows: (1) the proposal of HEFCSO algorithm, the novel metaheuristic for hyperparameter tuning in deep learning;”

Verdict: needs citation

Reason: This is an empirical claim about the proposal of a new algorithm (HEFCOS) in the study.

Sentence 18 · page unknown

“(2) a unified emotion detection pipeline that integrates pre-processing, GloVe embeddings, SMOTE-Text balancing, and the Bi-LSTM optical flow with attention optimized by HEFCOS; and (3) a thorough evaluation and benchmarking across multiple datasets and metrics against eight baselines on three publicly available datasets; statistical tests and interpretability are also illustrated.”

Verdict: needs citation

Reason: This is an empirical claim about a unified emotion detection pipeline and its evaluation, which requires citation.

Sentence 30 · page unknown

“To the best of our knowledge, authors are unaware of any previous work that hybridizes FA with CSA for hyperparameter optimization in BiLSTM emotion classification, making this the key novelty of this paper.”

Verdict: needs citation

Reason: This is an empirical claim about previous work in the field of hybridizing FA with CSA for hyperparameter optimization in BiLSTM emotion classification.

Sentence 31 · page unknown

“In the course of this study, three corpora from publicly available benchmarks were employed.”

Verdict: no citation needed

Reason: This is a general statement about the study methodology and does not require citation.

Sentence 35 · page unknown

“A particular-purpose six-phase pre-processing pipeline includes: 1) Dispensation of URLs, mentions, and hashtag marks.”

Verdict: no citation needed

Reason: This is a general statement about a pipeline structure and does not require citation.

Sentence 37 · page unknown

“Out-of-vocabulary tokens are initialized with character-level n-gram neighbour means.”

Verdict: needs citation

Reason: The sentence defines a term (out-of-vocabulary tokens) and implies a statistical or empirical claim about the initialization process, which requires citation.

Sentence 39 · page unknown

“The validation subset is used to assess the categorical cross-entropy objective function $J(?)$.”

Verdict: needs citation

Reason: The validation subset is used to assess a specific objective function in a machine learning context, which is an empirical claim.

Sentence 40 · page unknown

“Initialisation of 30 solutions is carried out randomly within search space ?.”

Verdict: needs citation

Reason: This is an empirical claim about the world. The use of 'randomly' implies a specific method of initialization.

Sentence 41 · page unknown

“Local Exploitation (Crow search algorithm phase): each crow m , starting randomly with a crow n , follows crow n to its memory M_n with awareness probability (AP) = 0.1.”

Verdict: needs citation

Reason: This is an empirical claim about the Crow search algorithm phase, which requires citation.

Sentence 42 · page unknown

“The switching of phases between different functions is governed by the physiologically motivated sigmoid probability switching probability, i.e., $p_{\text{switch}} = \frac{1}{1 + \exp(-(J_{\text{FA}} - J_{\text{CSA}}))}$.”

Verdict: needs citation

Reason: This is an empirical claim about the relationship between physiological factors and phase transitions in a system, which requires citation.

Sentence 43 · page unknown

“After 100 iterations, Q^* is determined to perform the final processing with BiLSTM.”

Verdict: needs citation

Reason: This is an empirical claim about the performance of a model (Q^*) after a certain number of iterations (100) using BiLSTM.

Sentence 44 · page unknown

“I carried out all the experiments on an Intel Core i7-11800H CPU, 16 GB RAM, NVIDIA GeForce RTX 3060 (6 GB)”

Verdict: needs citation

Reason: The sentence contains specific hardware details used in experiments, which is an empirical claim about the world.

Sentence 45 · page unknown

“RAM), Python 3.9, TensorFlow 2.10, and NLTK 3.7.”

Verdict: needs citation

Reason: The sentence lists specific software versions which are likely empirical claims about the world and require citation.

Sentence 46 · page unknown

“The data was stratified to train and validate/test at 70/15/15% of the dataset.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce new information that requires citation.

Sentence 47 · page unknown

“The evaluation metrics used in this work were accuracy, precision, recall, F1-score, the specificity, and Matthew correlation coefficient (MCC).”

Verdict: needs citation

Reason: The sentence describes evaluation metrics used in the work, which are empirical claims about the study's methodology.

Sentence 48 · page unknown

“HEFCSO-BiLSTM outperforms all eight baselines by achieving 96.42% accuracy and a 95.00% F1-score.”

Verdict: needs citation

Reason: The sentence claims a specific accuracy and F1-score for a model, which is an empirical claim about the model's performance.

Sentence 50 · page unknown

“Accuracy and F1-score comparisons across all methods are depicted in Fig. 2.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce new information that requires citation.

Sentence 51 · page unknown

“Per-emotion precision, recall, and F1-score for SemEval-2018 are presented in Table III.”

Verdict: needs citation

Reason: The sentence presents statistical data (precision, recall, F1-score) for a specific event (SemEval-2018) which is an empirical claim and requires citation.

Sentence 52 · page unknown

“The highest F1-scores are for joy (97.0%) and trust (96.0%).”

Verdict: needs citation

Reason: This is an empirical claim about F1-scores for specific emotions, which is high severity as it likely represents the author's own finding.

Sentence 53 · page unknown

“Disgust (93.1%) and fear (93.5%) are the most arduous due to their well-known ambiguous linguistic signal but significantly outperform their corresponding baseline.”

Verdict: needs citation

Reason: The sentence presents empirical claims about the prevalence of disgust and fear, which are percentages and thus require citation.

Sentence 54 · page unknown

“On the GoEmotions dataset, convergence curves of training loss were plotted over 50 epochs as illustrated by the graph above.”

Verdict: no citation needed

Reason: The sentence is stating a fact about the GoEmotions dataset and does not require citation as it is common knowledge.

Sentence 55 · page unknown

“HEFCSO-BiLSTM reaches a cross-entropy value of 0.072, which is significantly better than the values of 0.21 with FA-BiLSTM and 0.24 with CSA-BiLSTM, indicating the benefits of hybridization.”

Verdict: needs citation

Reason: The sentence contains a specific statistical value (cross-entropy value) that is compared to other values, indicating a claim about the effectiveness of a method.

Sentence 56 · page unknown

“Convergence is reached almost 28% faster.”

Verdict: needs citation

Reason: The sentence contains a specific statistic (28%) that likely requires citation as it is a claim about a specific result or finding.

Sentence 57 · page unknown

“BiLSTM, and ATAE-LSTM across a total of six evaluation”

Verdict: needs citation

Reason: BiLSTM and ATAE-LSTM are empirical claims about the world and the sentence refers to a total of six evaluation metrics or models.

Sentence 58 · page unknown

“HEFCSO-BiLSTM not only captures a significantly bigger area in all the dimensions but also shows extremely high tendency to significantly better performance, without any specific trade-offs coming into view.”

Verdict: needs citation

Reason: The sentence claims significant performance improvements without trade-offs, which is an empirical claim about the system's capabilities.

Sentence 59 · page unknown

“This presents paths where other predicted classes are misled.”

Verdict: needs citation

Reason: This is an empirical claim about the world, suggesting that other predicted classes are misled, which is a claim about the results or findings of a study.

Sentence 60 · page unknown

“The polycultural matrix sufficiently describes the diagonal dominance that shows high per-class accuracy.”

Verdict: needs citation

Reason: The sentence makes an empirical claim about the polycultural matrix describing diagonal dominance and per-class accuracy, which requires citation.

Sentence 61 · page unknown

“The highest misclassification was 8 instances between disgusted and angry, right exactly 7 between surprise and anticipation, which makes sense on the web from shared emotional valence across social media language contexts.”

Verdict: needs citation

Reason: The sentence presents empirical claims about misclassification rates and emotional valence, which are likely based on a study or research.

Sentence 62 · page unknown

“If HEFCSO is replaced with random initialization, the outcome goes down from 96.42% to 82.50%.”

Verdict: needs citation

Reason: The sentence presents a specific empirical claim about a statistical outcome, which requires citation.

Sentence 63 · page unknown

"If the attention layer is removed it costs 2.1 points; GloVe costs"

Verdict: needs citation

Reason: The sentence contains a specific numerical value (2.1 points) that likely refers to a study or experiment, indicating it requires citation.

Sentence 64 · page unknown

"3.3 points; SMOTE-Text costs 2.3 points."

Verdict: needs citation

Reason: The sentence contains a specific statistical claim (costs 2.3 points) that likely requires attribution as it is not common knowledge.

Sentence 65 · page unknown

"Using only FA or only CSA will set the benchmark low to 89.30% and 88.70%, respectively."

Verdict: needs citation

Reason: This is an empirical claim about specific benchmark percentages which are numbers and thus require citation.

Sentence 66 · page unknown

"The attention example for a representative sad-labelled tweet is shown in Fig. 8."

Verdict: no citation needed

Reason: The sentence is a simple description and does not require citation as it is common knowledge.

Sentence 67 · page unknown

"The model mainly focuses peak attention around 0.38 on the token 'devastated' and secondary attention about 0.18 on 'absolutely' while largely ignoring function words for attention (at around 0.07)."

Verdict: needs citation

Reason: The sentence presents specific attention values for particular tokens, which is an empirical claim about the model's behavior.

Sentence 68 · page unknown

“The high-attention tokens match entries from the NRC Emotion Lexicon on 91.3% of test samples, showing semantic alignment.”

Verdict: needs citation

Reason: The sentence is an empirical claim about semantic alignment using NRC Emotion Lexicon, which requires citation.

Sentence 69 · page unknown

“Cross-validation with a 10-fold showed confidence for statistical significance of all performance gains by paired t-tests ($p < 0.05$).”

Verdict: needs citation

Reason: Cross-validation and statistical significance are empirical claims that require citation as they present findings about the data.

Sentence 70 · page unknown

“The lowest t-statistics ($t = 4.82$ vis-a-vis the King-BiLSTM) exceed the standard value at $\alpha = 0.01$ with 9 degrees of freedom, which implies that improvements were not merely due to luck.”

Verdict: needs citation

Reason: The sentence presents a statistical claim about t-statistics and their significance level, which requires citation as it is an empirical claim.

Sentence 73 · page unknown

“In conclusion, the identification of attention (Fig. 8) proves that this model has been found to try to concentrate on tokens with significant emotional content, and at least makes the interpretation feasible for actual application.”

Verdict: no citation needed

Reason: The sentence is a conclusion and implies a finding that has been previously established in the context of the literature. The previous sentence, if present, would have carried the citation.

Sentence 74 · page unknown

“A major drawback of this framework is that it has been restricted to single-label problems.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce a new claim or result.

Sentence 75 · page unknown

“The computation cost (size of solving an optimization by the HEFCSO method is approximately 3.2 hours) has been growing in a heterogeneous scenario that suggests the integration with lightweight Transformers such as DistilBERT of the HEFCSO algorithm.”

Verdict: needs citation

Reason: The sentence contains an empirical claim about the computation cost of the HEFCSO method, which is a new finding and requires citation.

Sentence 77 · page unknown

“A major drawback of this framework is that it has been restricted to single-label problems.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce a new claim or result.

Sentence 78 · page unknown

“The computation cost (size of solving an optimization by the HEFCSO method is approximately 3.2 hours) has been growing in a heterogeneous scenario that suggests the integration with lightweight Transformers such as DistilBERT of the HEFCSO algorithm.”

Verdict: needs citation

Reason: The sentence contains an empirical claim about the computation cost of the HEFCSO method, which is a new finding and requires citation.

Sentence 80 · page unknown

“The paper discussed the development of HEFCSO-BiLSTM, a new model capable of recognizing emotions from social media text, combining a hybrid-nature inspired metaheuristic hyperparameter optimization by using BiLSTM with an attention mechanism, achieving state-of-the-art results in terms of 96.42% accuracy and an MCC of 0.545 on SemEval2018, with complete relative gains across all datasets.”

Verdict: needs citation

Reason: The sentence contains empirical claims about the model's performance and accuracy, which require citation.

Sentence 81 · page unknown

“Eight figures as a representation for comparison were made about system architecture (Fig. 1), accuracy comparison (Fig. 2), convergence curve (Fig. 3), radar chart (Fig. 4), confusion matrix (Fig. 5), cross dataset F1 (Fig. 6), ablation chart (Fig. 7), and attention heat map (Fig. 8).”

Verdict: no citation needed

Reason: The figures are common knowledge and do not require citation.

Sentence 82 · page unknown

“Future work should include: (i) performing fine-tuning of DistilBERT with the help of HEFCSO; (ii) moving to true multi-label emotion classification; (iii) extending for multilingual transfer to Hindi and code-switched text; and (iv) getting unified real-time streaming deployment on edge devices.”

Verdict: no citation needed

Reason: The sentence is a transition and does not introduce new information that requires citation.

Sentence 83 · page unknown

“The authors thank the Department of Computer Science & Engineering, DPG Institute of Technology and Management, Gurgaon, and the research community that curated and openly released the SemEval-2018, ISEAR, and Go-Emotions datasets used in this work.”

Verdict: no citation needed

Reason: The sentence is a standard acknowledgment of support and resources used in the work, which is considered common knowledge.

Cited, but not checkable

These sentences DO cite something. Gaply could not check them because the cited work is not available to it — which is a gap in the library, not a defect in the writing.

Sentence 6 · page unknown

“Exact profiling of microemotional states from such text carries points to mental health monitoring [1], crisis detection [2], customer sentiment analytics [3], and public opinion modeling [4].”

Reason: [4] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 8 · page unknown

“Classical lexical-based approaches [5] or superficial learning classifiers [6] suffer from a lack of good reasoning and understanding about out-of-vocabulary words and domain shift.”

Reason: [6] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 10 · page unknown

“[7] does it with a fair degree of success when it works best at capturing full chronological context in the document at surface structure.”

Reason: [7] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 19 · page unknown

“Initially, emotion detection systems relied on decision trees and lexical resources such as SentiWordNet [5] and the NRC Emotion Lexicon [11] before classifiers were considered suitable.”

Reason: [11] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 20 · page unknown

“Alm et al. [12] proposed rule-based techniques which fell out of favor in a time when Mohammad and Turney [15] were actively engaged in lexicon expansion so as to accommodate tweets along with a number of design limitations and incomplete interpretations, these Twitter-related lexicons fail to account for the context-dependent polarity reversal or blend in with the highly informal lingo prevalent in social media”

Reason: [15] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 21 · page unknown

“LSTM units (Schmidhube 1997) have tackled the problem of the vanishing gradient in RNNs (Elman 1990).”

Reason: cited work not in library: (Elman 1990)

- What to do: Add the cited work to your library so this sentence can be checked against it.

Sentence 22 · page unknown

“The TD-LSTM was introduced by Tang et al. (2016) to model aspects of the sentiment.”

Reason: cited work not in library: Tang et al. (2016)

- What to do: Add the cited work to your library so this sentence can be checked against it.

Sentence 23 · page unknown

“The ATAELSTM model was proposed by Wang et al. (2016) to incorporate the aspect information into the attention gate settings.”

Reason: cited work not in library: Wang et al. (2016)

- What to do: Add the cited work to your library so this sentence can be checked against it.

Sentence 24 · page unknown

“The Bidirectional LSTM extensions (Graves and Schmidhuber 2005) have shown efficiencies across SemEval compared to unidirectional models.”

Reason: cited work not in library: (Graves and Schmidhuber 2005)

- What to do: Add the cited work to your library so this sentence can be checked against it.

Sentence 25 · page unknown

"In contrast, GoEmotions by Demszky et al. (2020) fine-tuned BERT and reached only 46% macro-F1, over 27 categories, which is substantially lower than the 95% macro-F1 of the present study."

Reason: cited work not in library: Demszky et al. (2020)

- What to do: Add the cited work to your library so this sentence can be checked against it.

Sentence 26 · page unknown

"Recurrent network parameters have been tuned by PSO [19] and Genetic Algorithms [8]."

Reason: [8] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 27 · page unknown

"Yang [9] formalized FA by describing a light intensity attenuation mechanism."

Reason: [9] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 28 · page unknown

"CSA was primarily introduced for continuous optimization purposes by Askarzadeh [10]."

Reason: [10] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 29 · page unknown

"Hybridizations strategies showed enhancement by synergy [20]."

Reason: [20] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 32 · page unknown

"For instance, the SemEval-2018 Task 1 dataset [21] consists of 10,983 English tweets that are labeled with eight Plutchik emotions."

Reason: [21] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 33 · page unknown

"The ISEAR dataset [22] consists of 7,666 self-reported emotional statements."

Reason: [22] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 34 · page unknown

"Finally, the GoEmotions dataset [18] consists of 58,009 English Reddit comments labeled with 27 fine-grained emotions that were remapped into eight categories."

Reason: [18] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Sentence 36 · page unknown

"Each token is mapped to a 200-dimensional GloVe vector pre-trained on two billion tweets [24]."

Reason: [24] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer — not in your library

- What to do: Add this work to your library — "Fetch open-access PDF" will try to find a free copy by DOI, or attach a PDF you already have.

Sentence 49 · page unknown

"Its closest rival, King-BiLSTM [10], reached only 74.90% specificity, therefore showing some class-specific bias."

Reason: [10] is not in the reference list

- What to do: This number has no entry in the paper's own reference list. Check the citation is numbered correctly.

Not judged

The model's answer for these did not pass Gaply's own checks, twice. They are reported rather than hidden: an unanswered sentence is not a clean one.

Sentence 38 · page unknown

“HEFCSO considers each configuration as a 4-dimensional vector $\mathbf{v} = [n_units, lr, dropout, batch_size]$.”

Reason: the model's output failed validation twice: search_query: must be null when needs_citation is false

Sentence 71 · page unknown

“The main difference emerges in the method of switching different strategies and combined optimization within the HEFCSO approach.”

Reason: the model's output failed validation twice: search_query: must be null when needs_citation is false

Sentence 72 · page unknown

“FA explores very quickly in terms of solutions not related to the problem, which in the beginning avoids convergence prematurely, while CSA goes on refining solution within substantially better spots and at a rate they begin to group together.”

Reason: the model's output failed validation twice: search_query: must be null when needs_citation is false

Sentence 76 · page unknown

“Using entirely free and open-source tools to create the method demonstrates how the approach can be readily incorporated for use in academic research settings.”

Reason: the model's output failed validation twice: search_query: must be null when needs_citation is false

Sentence 79 · page unknown

“Using entirely free and open-source tools to create the method demonstrates how the approach can be readily incorporated for use in academic research settings.”

Reason: the model's output failed validation twice: search_query: must be null when needs_citation is false

Some characters could not be displayed in this report and appear as a question mark. This is a limitation of the report's font, not of the analysis — nothing was skipped or removed.